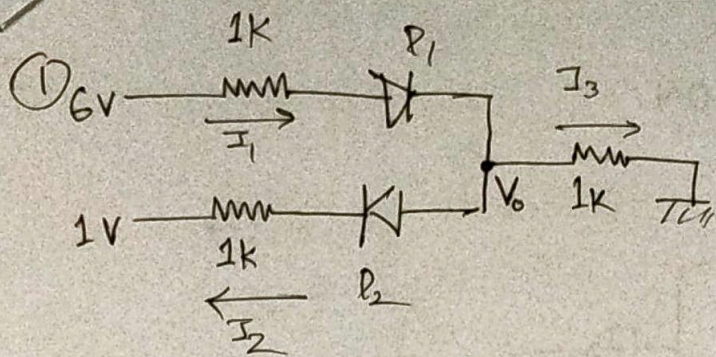
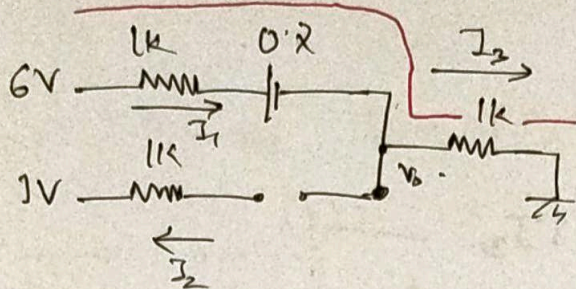




Find I_1, I_2, I_3, V_0 . Use CVD model with $V_D = 0.7V$

Solⁿ: Assume $D_1 = ON, D_2 = OFF$



$$I_2 = 0 \quad [OFF]$$

$$I_1 = I_2 + I_3 \quad [KCL]$$

$$\therefore I_1 = I_3$$

Writing KVL along 6V to GND [Red line]

$$6 = 1I_1 + 0.7 + 1I_3 + 0$$

$$\Rightarrow 2I_1 = 6 - 0.7 \quad [since \ I_1 = I_3]$$

$$\Rightarrow I_1 = I_3 = 2.65 \text{ mA}$$

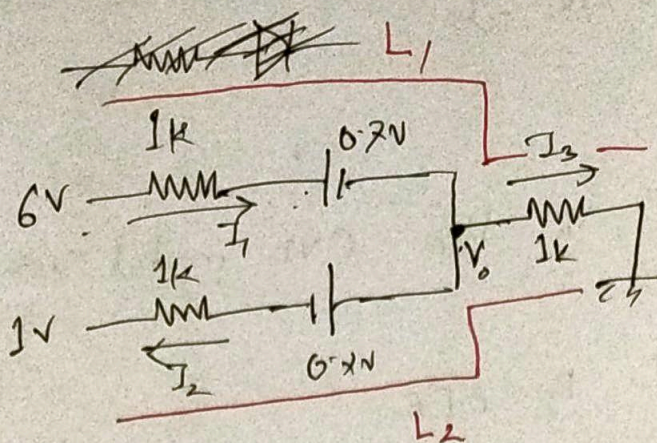
$$\therefore V_0 = 1I_3 = 2.65V$$

Now $I_1 = +ve$, so D_1 assumption correct

But for D_2 ,
 $V_A = V_0 = 2.65V$
 $V_C = 1V$
 $\therefore V_D = 1.65V$

So assumption wrong

Assume $D_1 = ON, D_2 = ON$



KVL along L_1

$$6 = 1I_1 + 0.2 + 1I_3 \quad \text{--- (i)}$$

KVL along L_2

$$1 = -1I_2 - 0.2 + 1I_3 \quad \text{--- (ii)}$$

KCL $\Rightarrow$

$$I_1 = I_2 + I_3 \quad \text{--- (iii)}$$

Solving (i), (ii) and (iii)

$$I_1 = \cancel{1.3 \text{ mA}} \quad 2.96 \text{ mA}$$

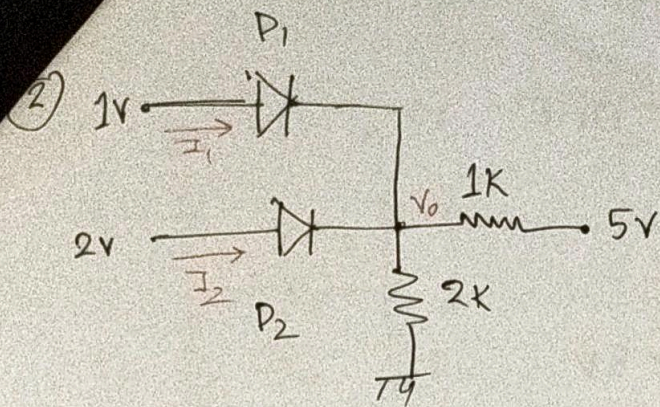
$$I_2 = 0.63 \text{ mA}$$

$$I_3 = 2.33 \text{ mA}$$

$$\therefore V_0 = 1I_3 = 2.33 \text{ V}$$

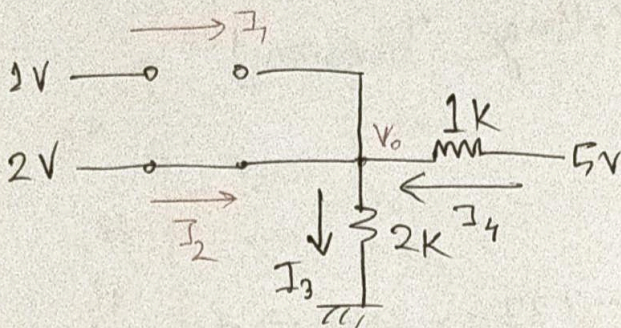
$I_1 = +ve \Rightarrow D_1$ assumption correct ✓

$I_2 = +ve \Rightarrow D_2$ ✓



Find I_1 , I_2 and V_0 . Assume the diodes are ideal.

Solⁿ: Assume $D_1 = \text{OFF}$, $D_2 = \text{ON}$



$$\therefore I_1 = 0 \text{ (OFF)}$$

$$\therefore V_0 = 2V$$

$$\therefore I_3 = \frac{V_0}{2} = 1\text{mA}, \quad I_4 = \frac{5 - V_0}{1} = \frac{5 - 2}{1} = 3\text{mA}$$

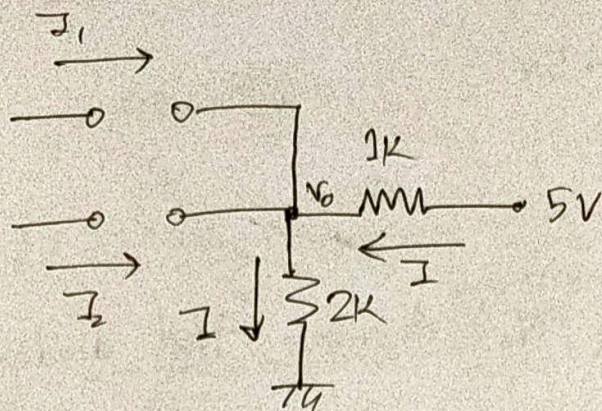
$$\text{KCL} \Rightarrow I_2 + I_4 = I_3 \Rightarrow I_2 = I_3 - I_4$$

$$\therefore \boxed{I_2 = -2\text{mA}}$$

$\therefore D_2$ assumption wrong

For D_1 , $V_D = V_A - V_C = 1 - 2 = -1 \Rightarrow$ Assumption correct

Assume $D_1 = \text{OFF}$, $D_2 = \text{OFF}$



$$I_1 = 0$$

$$I_2 = 0$$

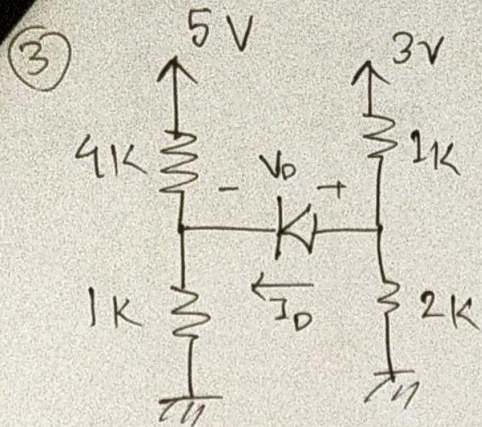
$$V_o = \frac{2}{2+1} \times 5 \quad [\text{Voltage Division}]$$

$$\Rightarrow V_o = 3.33V$$

For $D_1 \Rightarrow V_D = V_A - V_C = 1 - 3.33 = -2.33 \Rightarrow \text{Assumption correct}$

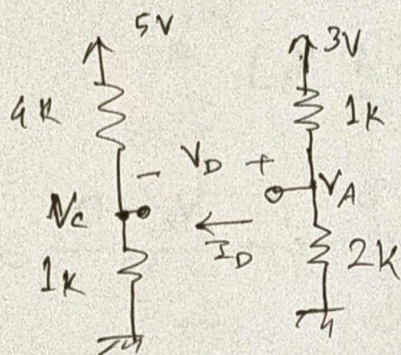
For $D_2 \Rightarrow V_D = V_A - V_C = 2 - 3.33 = -1.33 \Rightarrow$

✓



Find V_D and I_D . Use CVD with $V_D = 0.7V$.

Solⁿ: Assume diode OFF



$$\therefore I_D = 0$$

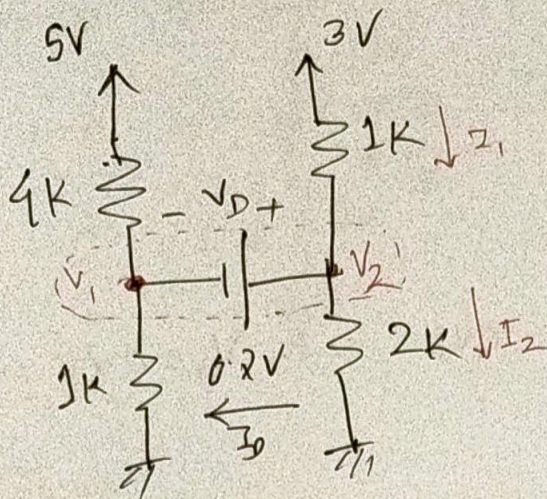
$$V_A = \frac{2}{2+1} \times 3 = 2V \quad [\text{voltage division}]$$

$$V_C = \frac{1}{4+1} \times 5 = 1V \quad [\quad \sim \quad \sim]$$

$$\therefore V_D = V_A - V_C = 2 - 1 = 1V$$

$\therefore$ Assumption wrong!

So, the diode would be on!



$$\therefore V_D = 0.2V$$

Super node equation (V_1 and V_2)

$$\frac{V_1 - 5}{4} + \frac{V_1 - 0}{1} + \frac{V_2 - 3}{1} + \frac{V_2 - 0}{2} = 0 \quad (i)$$

$$V_2 - V_1 = 0.2 \quad (ii)$$

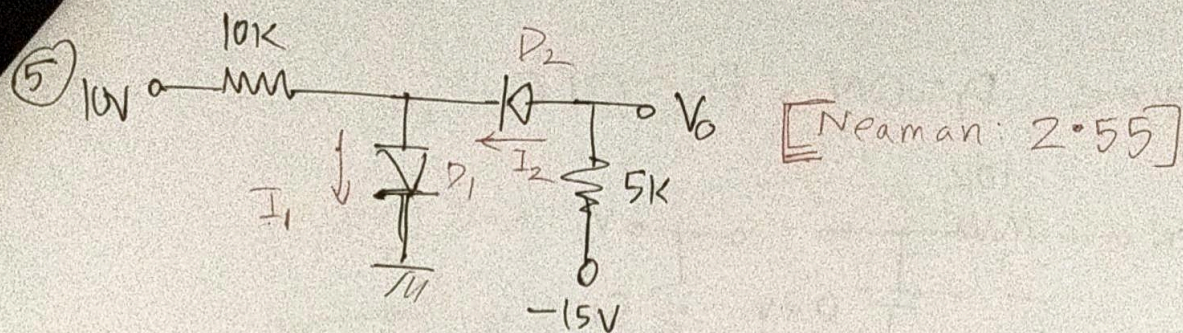
solving (i) and (ii)

$$V_1 = \cancel{1.636V} \quad 1.63V$$

$$V_2 = 1.863V$$

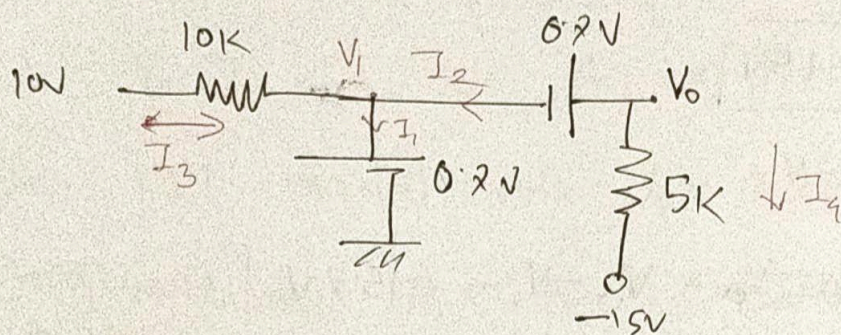
$$I_1 = \frac{3 - V_2}{1} = 1.137 \text{ mA}, \quad I_2 = \frac{V_2 - 0}{2} = 0.932 \text{ mA}$$

$$\therefore I_D = I_1 - I_2 \text{ [KCL]} = 0.205 \text{ mA [correct assumption]}$$



Find I_1, I_2
 V_0 . Use CVD model with $V_D = 0.7V$.

Soln: Assume $D_1 = ON, D_2 = ON$.



$$V_1 = 0.7V$$

$$V_0 = V_1 + 0.7V = 1.4V$$

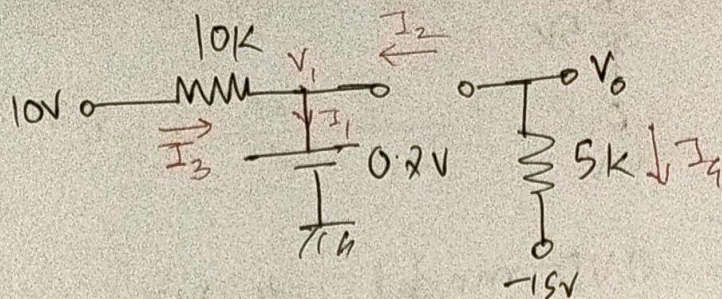
$$I_3 = \frac{10 - 0.7}{10} = 0.93mA$$

$$I_4 = \frac{1.4 - (-15)}{5} = 3.28mA$$

From KCL, $I_2 = -I_4 = -3.28mA$ [Assumption wrong for D_2]

$I_1 = I_3 + I_2 = -2.35mA$ [Assumption wrong for D_1]

Assume $D_1 = \text{ON}$, $D_2 = \text{OFF}$



Open CKT $\Rightarrow I_2 = 0$

$I_4 = 0$

$\therefore \boxed{V_0 = -15V}$

$V_1 = 0.7V$

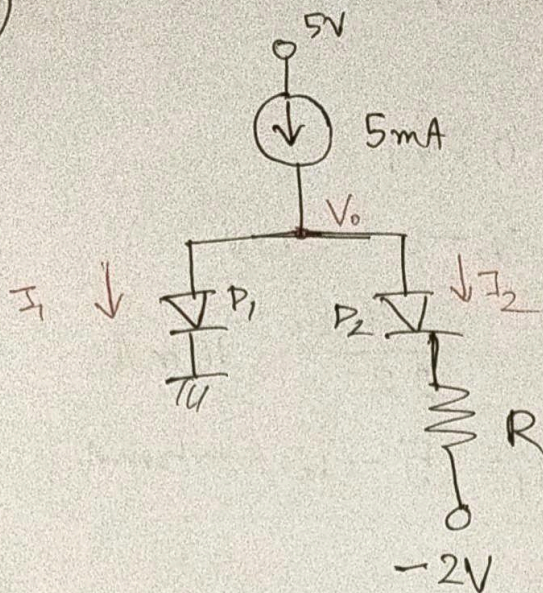
$\therefore V_{D2} = V_0 - V_1 = V_0 - 0.7 = -15.7V$ [Assumption correct for D_2]

$I_3 = \frac{10 - V_1}{10} = \frac{10 - 0.7}{10} = 0.93mA$

$I_1 = I_3 = 0.93mA$ [KCL] [Assumption correct for D_1]

So, $\boxed{V_0 = -15V}$ Ans

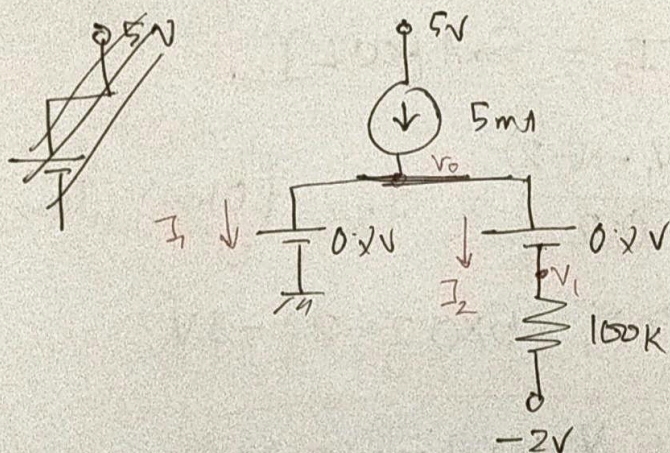
⑥



Find V_0 , I_1 and I_2 for a) $R = 100\text{K}$, b) $R = 0.2\text{K}$.

Use CVD model with $V_f = 0.2\text{V}$.

Solⁿ: a) assume $D_1 = \text{ON}$, $D_2 = \text{ON}$



$$V_0 = 0.2\text{V}$$

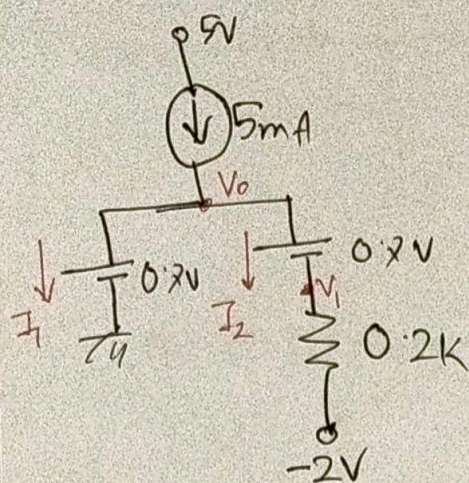
$$V_1 = V_0 - 0.2 = 0\text{V}$$

$$\therefore I_2 = \frac{0 - (-2)}{100} = 0.02\text{mA}$$

$$I_1 = 5 - I_2 \text{ [KCL]} = 4.98\text{mA}$$

Since I_1 and I_2 both are positive, Assumption correct.

⑥ Assume $D_1 = \text{ON}, D_2 = \text{ON}$



$$V_0 = 0.2V$$

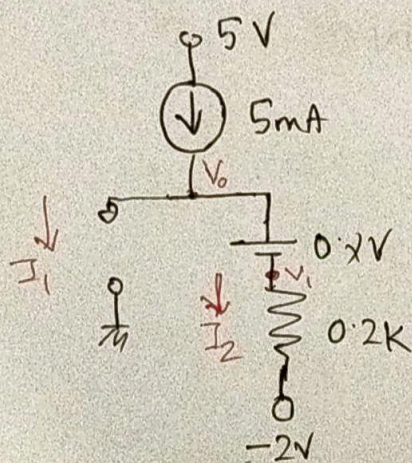
$$V_1 = V_0 - 0.2 = 0V$$

$$I_2 = \frac{0 - (-2)}{0.2} = 10mA$$

$$\therefore I_1 = 5 - I_2 = -5mA$$

Since current through D_1 is negative, assumption wrong.

Assume $D_1 = \text{OFF}, D_2 = \text{ON}$



$$I_1 = 0mA$$

$$I_2 = 5mA \text{ [KCL]}$$

$$\frac{V_1 - (-2)}{0.2} = I_2 \text{ [Ohm's law]}$$

$$\Rightarrow V_1 = 5 \times 0.2 - 2 = -1V$$

$$V_0 = V_1 + 0.2 = -0.3V$$

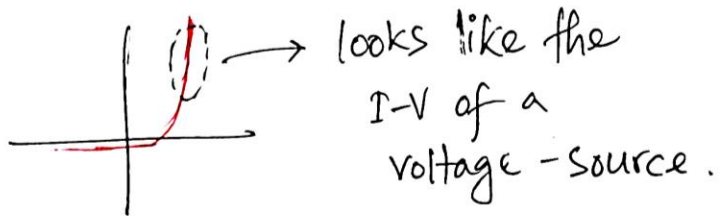
Since $V_D = -0.3V$ for D_1 and $V_D < 0.2V$,

assumption for D_1 correct.

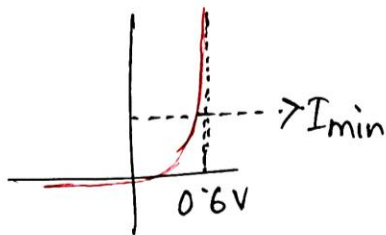
$I_1 = +ve \Rightarrow$ Assumption for D_2 correct.

Applications:

2. Voltage Regulation:

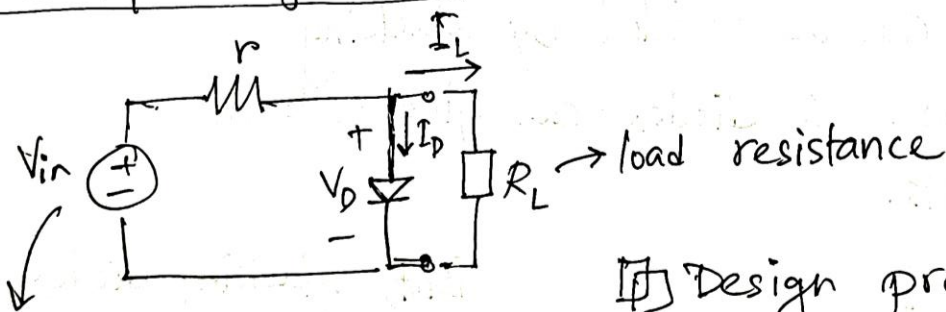


We can use the forward-drop of a non-ideal diode to give a constant voltage at the output.

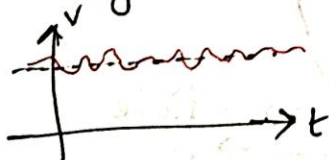


However, the vertical I-V is maintained only if the diode-current is greater than I_{min} .

Example regulator:



unregulated voltage source



↳ voltage is not perfect DC
- has ripples/
fluctuation

Design problem:

$$V_{in} = 3V \pm 0.1V$$

$$I_{min} = 1mA$$

Load can draw max^m 10mA current
What will be r for worst-case?
if V_D is to be regulated at 0.6V?

For worst case,

Load current is max^m, i.e., $I_L = 10mA$

Diode " " min^m, i.e., $I_D = I_{min} = 1mA$

Input voltage " " , i.e., $V_{in} = 3V - 0.1V = 2.9V$

∴ To maintain regulation at worst-case,

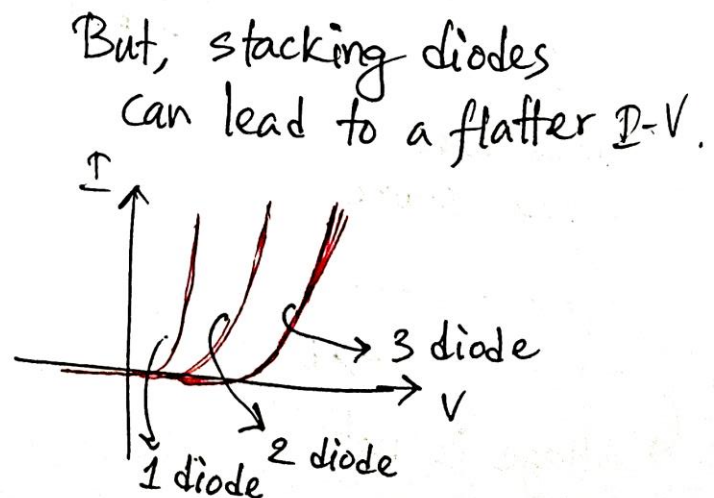
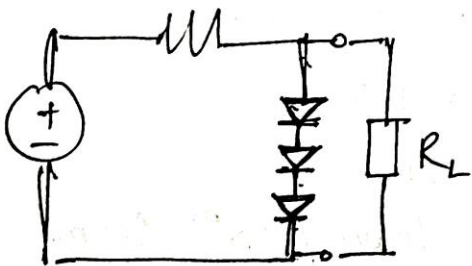
$$\frac{V_{in(min)} - 0.6}{r} = I_L + I_D$$

$$\Rightarrow r = \frac{(2.9 - 0.6)V}{10mA + 1mA}$$
$$\cong \cancel{2000} \ 209\Omega$$

Drawbacks of regulator

1. Regulation voltage is low (only 0.6V!!)
2. I_{min} is relatively high
3. r can become low, and, cause high power loss.

— First drawback can be avoided by stacking diodes in series. 3 diodes can give voltage close to 2 Volts.

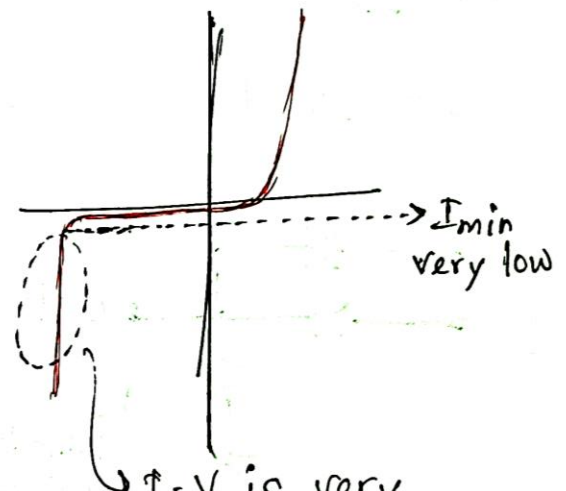


★ So, we usually don't stack more than 3 diodes.

- Better solⁿ ???

⇒ Use the reverse-breakdown region of the diode I-V.

- Pros
1. Breakdown voltage can be controlled during fabrication.
 2. I_{min} for reverse-breakdown can be very low.



→ I-V is very vertical during breakdown.

Very good voltage source!!
(☺)

Diodes which are designed/fabricated to operate reliably in breakdown are Zener Diodes.

Symbol:

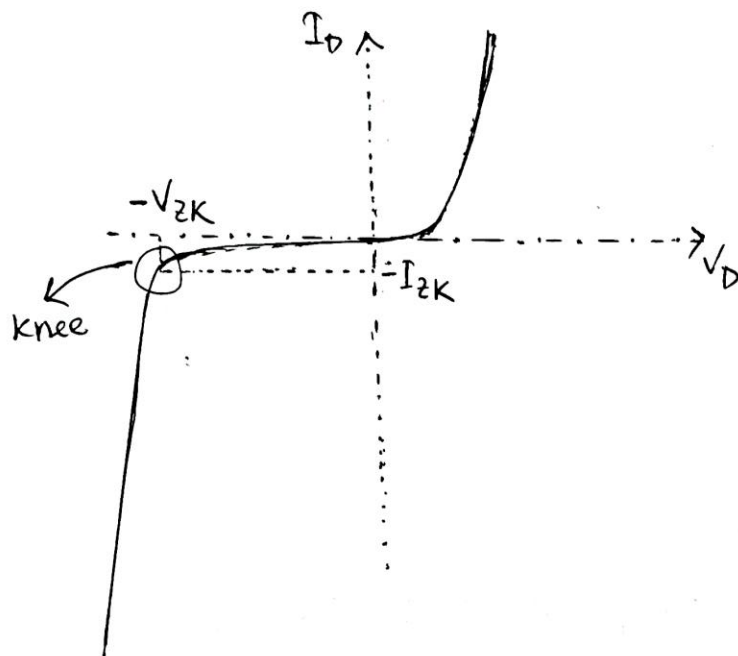
Practice Problem:

Design a voltage regulator using a 2V-Zener Diode with the constraints of the previous problem.

Find the value of r in this case.

Voltage Regulators with Non-Ideal Zeners

Previously, we assumed that, the IV of a zener-diode is completely vertical in breakdown. However, for a real zener diode in breakdown, there is some slope in the IV.



real

So, during reverse breakdown, a ^{**real**} zener-diode may be modelled as a voltage source with a series resistance.

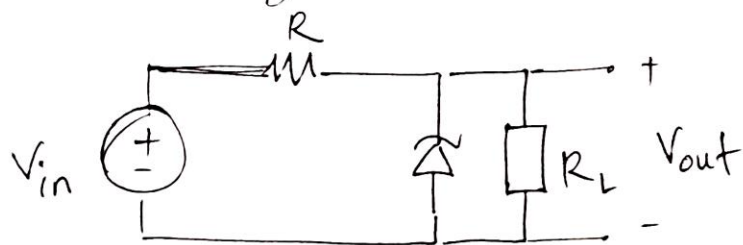
The minimum point upto which the zener can stay in breakdown is called the knee. The voltage and current at that point are the knee-voltage (V_{zk}) and knee-current (I_{zk}).

To stay in break-down, the reverse zener current must be at least, or, more than the knee-current.

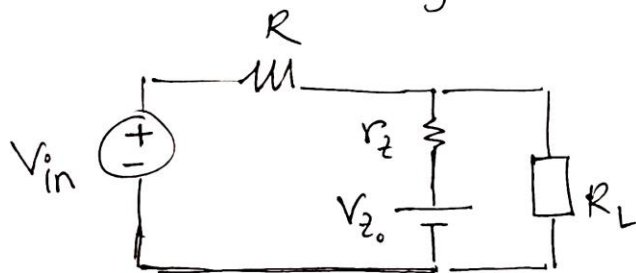
* In spite of having a slope in break-down, the IV is almost vertical. So, the series resistance is very small. Usually, it is in the range $10\Omega \sim 50\Omega$.

Characteristics of a Non-ideal Voltage Regulator:

Our basic regulator ckt is this:



When considering a reverse IV with a slope, we can replace it with a Voltage Source and Series resistance.



[Note that... $V_{Z0} \neq V_{ZK}$. Although, they are pretty close.]

For a regulator like this, the output voltage doesn't stay constant if the Input Voltage or the Output Current changes. V_{out} changes. We want to quantify this change.

2 measures of this change are:

(1) Line Regulation:

This is the change in output voltage with respect to the change in line voltage (i.e, Input Voltage). — keeping load resistance fixed.

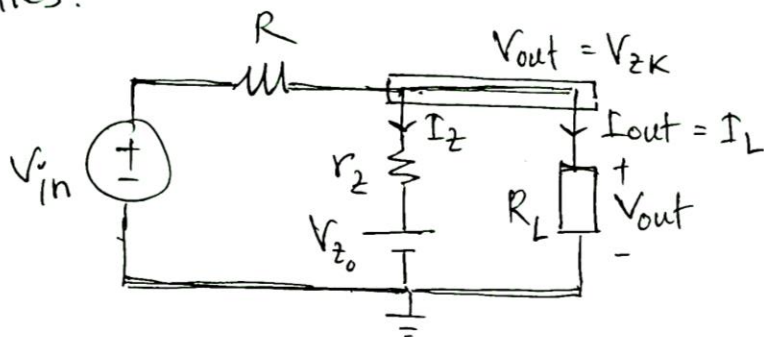
Mathematically, line regulation = $\frac{dV_{out}}{dV_{in}}$.

(2) Load Regulation:

This is the change in output voltage w.r.t. load current (i.e., output current.) — keeping input voltage fixed.

Mathematically, load regulation = $\frac{dV_{out}}{dI_{out}}$

For our basic voltage regulator, we will find these 2 quantities.



As before we will proceed from the node equation of V_{out} .

$$\frac{V_{out} - V_{in}}{R} + I_2 + I_L = 0 \Rightarrow \frac{V_{out} - V_{in}}{R} + \frac{V_{out} - V_{z0}}{r_2} + I_{out} = 0 \quad \text{--- (1)}$$

For line-regulation, $I_{out} = \frac{V_{out}}{R_L}$.

$$\therefore \frac{V_{out} - V_{in}}{R} + \frac{V_{out} - V_{z0}}{r_2} + \frac{V_{out}}{R_L} = 0.$$

$$\Rightarrow V_{out} \left(\frac{1}{R} + \frac{1}{r_2} + \frac{1}{R_L} \right) = \frac{1}{R} V_{in} + \frac{1}{r_2} V_{z0}$$

$$\therefore V_{out} = \frac{R \parallel r_2 \parallel R_L}{R} \cdot V_{in} + \frac{R \parallel r_2 \parallel R_L}{r_2} \cdot V_{z0} \quad \text{--- (11)}$$

Here, $R_p = R \parallel r_2 \parallel R_L$ is the parallel combination of R , r_2 and R_L .

$$\text{i.e., } \frac{1}{R_p} = \frac{1}{R} + \frac{1}{r_2} + \frac{1}{R_L}$$

$$\therefore \text{From (11), } \frac{dV_{out}}{dV_{in}} = \frac{R \parallel r_2 \parallel R_L}{R} \quad [\text{Line Regulation}] \quad \text{--- (3)}$$

As a special case, if no load is connected to the output, (i.e., if output is open-circuited or $R_L = \infty$)

$$\frac{dV_{out}}{dV_{in}} = \frac{R \parallel r_z}{R} = \frac{r_z}{r_z + R} \quad \text{--- ()}$$

For load-regulation, we can start directly from eqn (1),

$$\frac{V_{out} - V_{in}}{R} + \frac{V_{out} - V_{z_0}}{r_z} + I_{out} = 0.$$

$$\Rightarrow V_{out} \left(\frac{1}{R} + \frac{1}{r_z} \right) = -I_{out} + \frac{1}{R} V_{in} + \frac{1}{r_z} V_{z_0}.$$

$$\therefore V_{out} = - (R \parallel r_z) \cdot I_{out} + \frac{R \parallel r_z}{R} V_{in} + \frac{R \parallel r_z}{r_z} V_{z_0}.$$

As before, $R_p = R \parallel r_z$ is the parallel combⁿ of R and r_z .

$$\text{i.e., } \frac{1}{R_p} = \frac{1}{R} + \frac{1}{r_z}.$$

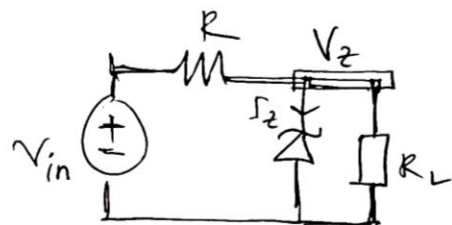
$$\therefore \frac{dV_{out}}{dI_{out}} = - R \parallel r_z = - \frac{R \cdot r_z}{R + r_z} \quad [\text{Load Regulation}] \text{ --- (4)}$$

In both cases, eqn (3) and (4), the regulation decreases with r_z . For a Zener with completely vertical IV, $r_z = 0$, and, both $\frac{dV_{out}}{dV_{in}}$ and $\frac{dV_{out}}{dI_{out}}$ will be equal to 0.

Finding corner parameters with r_z :

To find the corner values of different parameters of voltage regulator $[V_{in}(\min), I_L(\max), R_L(\min), R(\max)]$, we use the following node-equation:

$$\frac{V_z - V_{in}}{R} + I_z + I_L = 0$$



To find the corner values, we always have to set,

$$V_z = V_{zk} \text{ and } I_z = I_{zk}.$$

If one of them is not given, they have to be inferred from other data.

E.g, if V_{z0} , r_z and I_{zk} are given, we can find V_{zk} using,

$$V_{zk} = V_{z0} + I_{zk} \cdot r_z$$

Example:

You are a circuit-engineer who has to design a voltage regulator. The zener diodes you ~~have~~ own have a knee current of 1mA, a series resistance of 25Ω and $V_{z0} = 4V$.

The customer to which you will sell the regulator has a supply of 6V with $\pm 0.2V$ ripple. ~~he~~ The lowest resistance that he will connect to the output of the regulator is $0.5k\Omega$.

Find the value of R such that the regulator will work properly in the worst-case condition.

Here, we will use the ~~one~~ worst-case values for all parameters.

$$I_{ZK} = I_Z = I_{ZK} = 1 \text{ mA}$$

$$\begin{aligned} V_Z &= V_{ZK} = V_{Z_0} + I_{ZK} \times r_Z \\ &= 4 + 1 \times 0.025 \quad [r_Z = 25 \Omega = 0.025 \text{ k}\Omega] \\ &= 4.025 \text{ V} \end{aligned}$$

[Note: If V_{ZK} was given directly in the question, we wouldn't have to do this step].

$$I_L = I_L(\text{max}) = \frac{V_Z}{R_L(\text{min})} = \frac{V_{ZK}}{R_L(\text{min})} = \frac{4.025}{0.5} = 8.05 \text{ mA}$$

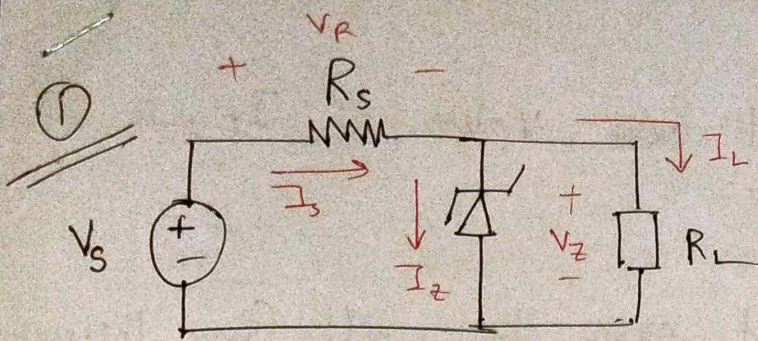
$$V_{in} = V_{in}(\text{min}) = 6 \text{ V} - 0.2 \text{ V} = 5.8 \text{ V}.$$

Plugging all these values in the eqn—

$$\frac{V_Z - V_{in}}{R} + I_Z + I_L = 0.$$

$$\text{We get, } R(\text{max}) \approx 0.196 \text{ k}\Omega \approx 200 \Omega.$$

- ** If V_{ZK} , V_{Z_0} and r_Z ^{are given,} we can calculate I_{ZK} from that.
- ** If both V_{Z_0} and r_Z are not given, only the zener breakdown voltage is given, we can use $V_{ZK} = V_Z$ (breakdown).
- ** If none of V_{Z_0} , r_Z , I_{ZK} are given, we can use $I_{ZK} = 0$.



The Zener diode is specified as

$$V_{z0} = 3V$$

$$P_z = 0.5W$$

$$I_{zk} = 1mA$$

The load current can vary from $0mA$ to $50mA$.

Ⓐ The source has a nominal value of $5V$ and can vary $\pm 10\%$.

(a) Find $V_s(\min)$, $V_s(\max)$, $I_L(\min)$, $I_L(\max)$

Ⓑ Solⁿ: $10\% \text{ of } 5V = 0.5V$

$$\therefore V_s(\min) = 5 - 0.5 = 4.5V$$

$$V_s(\max) = 5 + 0.5 = 5.5V$$

$$I_L(\min) = 0 \text{ mA}$$

$$I_L(\max) = 50mA$$

(b) What is the worst case values of I_L , I_Z and V_S ?

Solⁿ: In worst case, I_{ZK} will flow through the Zener diode. [If the current falls below this, the regulation will not be maintained]

$$KCL \Rightarrow I_Z = I_S - I_L$$

$$\text{and } I_S = \frac{V_S - V_L}{R_S}$$

So, I_Z is minimum when I_L is maximum and V_S is minimum.

$\therefore$ Worst case $\Rightarrow$

$$I_L = I_L(\text{max}) = 50 \text{ mA}$$

$$I_Z = I_Z(\text{min}) = I_{ZK} = 1 \text{ mA}$$

$$V_S = V_S(\text{min}) = 4.5 \text{ V}$$

(c) For worst case, what is the value of I_S ?

Solⁿ: $I_S = I_Z + I_L = 50 + 1 = 51 \text{ mA}$

(d) ~~For worst case,~~

(d) For worst case, what is the value of V_o ?

Solⁿ: In worst case scenario, the Zener diode will barely maintain its voltage.

~~So $V_L = V_Z$~~

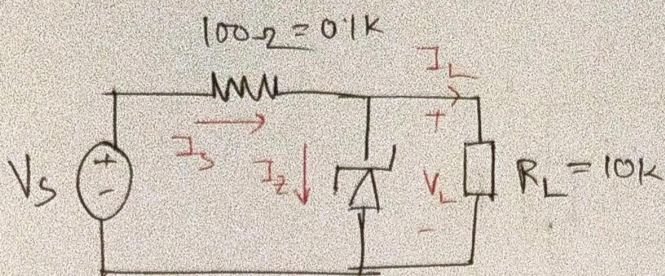
So, $V_L = V_Z = V_Z + I_Z r_Z = 3V$.

$$\begin{aligned}\therefore V_o &= V_s - V_L \\ &= 4.5 - 3 \\ &= 1.5V\end{aligned}$$

(e) Find the value of R_s , for which the Zener diode maintains regulation in worst case scenario.

Solⁿ: $R_s = \frac{V_s}{I_Z} = \frac{1.5}{51} \approx 30\Omega$.

2



Zener diode specification: $V_{Z0} = 3V$

$$r_z = 0.2$$

$$I_{ZK} = 1mA$$

Find the minimum value of V_s for which the Zener diode maintain its regulation.

Solⁿ: In worst case scenario, $I_z = I_{ZK} = 1mA$.

$$\therefore V_L = V_z = V_{Z0} + I_z r_z = 3V$$

$$\therefore I_L = \frac{V_L}{R_L} = \frac{3V}{10K\Omega} = 0.3mA$$

$$\therefore I_s = I_z + I_L \quad (\text{KCL})$$

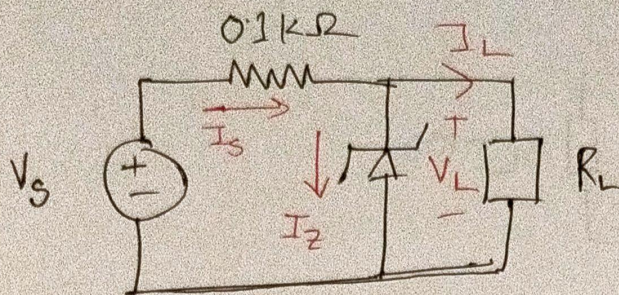
$$= 1.3mA$$

$$I_s = \frac{V_s - V_L}{0.1K} = 1.3mA$$

$$\Rightarrow \frac{V_s - 3}{0.1K} = 1.3mA$$

$$\Rightarrow \boxed{V_s = 3.13V} \quad \therefore \boxed{V_s(\text{min}) = 3.13V}$$

③



Zener diode specification: $V_Z = 3V$,

$$r_Z = 0.2$$

$$I_{ZK} = 1mA$$

The source V_s has a nominal value of 5V and can vary $\pm 10\%$.

Find the worst case/minimum value of R_L so that regulation is maintained.

Solⁿ: In worst case, $I_Z = I_{Z(min)} = I_{ZK} = 1mA$

$$V_L = V_Z = V_Z + I_Z r_Z = 3V$$

In worst case, $V_s = V_{s(min)} = 5 - 10\% = 5 - 0.5 = 4.5V$

$$\therefore I_s = \frac{V_s - V_L}{0.1k} = \frac{4.5 - 3}{0.1k} = 15mA$$

$$\begin{aligned} \therefore I_L &= I_s - I_Z \text{ [KCL]} \\ &= 15 - 1 \\ &= 14mA \end{aligned}$$

$$\therefore R_L = \frac{V_L}{I_L} = \frac{3V}{14mA} \approx 214\Omega$$